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ORIGINAL RESEARCH

On the structure of cyclic codes over $M_2(\mathbb{F}_p + u\,\mathbb{F}_p)$

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Abstract In this article, for a prime p such that $p \equiv 2$ or $3 (\mathrm{mod}\ 5)$, we identify cyclic codes of length N over $R = M_2(\mathbb{F}_p + u\mathbb{F}_p)$, $u^2 = 0$ as right R-submodules of $R[x]/\langle x^N - 1\rangle$. Also, we define an isometry from $M_2(\mathbb{F}_p + u\mathbb{F}_p)$ to $\mathbb{F}_{p^2} + u\mathbb{F}_{p^2} + v\mathbb{F}_{p^2} + uv\mathbb{F}_{p^2}$, where $u^2 = v^2 = 0, uv = vu$ and determine the structure of cyclic codes, in particular self-dual cyclic codes of length N where $\gcd(N, p) = 1$. Moreover, several optimal and near to optimal codes are obtained as the Gray images of these codes over R.

Keywords Cyclic code · Gray map · Self-dual code · Matrix ring · Optimal code

Mathematics Subject Classification 94B05 · 94B15 · 94B35 · 94B60

1 Introduction

The class of cyclic codes is an important class of linear codes which has been extensively studied over finite rings after the establishing work of Hammons et al. [7] in 1994. Like commutative rings [9, 11–14], noncommutative rings also play an important role in the development of error-correcting codes. Apart from skew polynomial rings [4, 8, 10], rings of matrices have recently attracted a growing interest among researchers. In 2012, Oggier et al. [16] considered the matrix ring $M_2(\mathbb{F}_2)$ for space-time coding. In 2013, Alahmadi et al. [1] studied cyclic codes over $M_2(\mathbb{F}_2)$ and explored certain self-dual $\mathbb{F}_4$-codes. Also, in 2014, Falcunit and Sison [5] investigated the structure of cyclic codes over $M_2(\mathbb{F}_p)$ by using an isometry between $M_2(\mathbb{F}_p)$ and $\mathbb{F}_{p^2} + u\mathbb{F}_{p^2}$. Recently, Luo and Udaya [15] considered the matrix ring $M_2(\mathbb{F}_2 + u\mathbb{F}_2)$ and obtained several optimal $\mathbb{F}_4$-codes of length 16 and 20, respectively as the Gray images of cyclic codes. Further, Pal et al. [17] and Bhowmick et al. [2] have studied cyclic codes over $M_4(\mathbb{F}_2)$ and self-dual cyclic codes over $M_2(\mathbb{Z}_4)$, respectively.

In this paper, for any integer $k \geq 0$, we consider the matrix ring $M_2(\mathbb{F}_p + u\mathbb{F}_p)$ where $u^2 = 0$ and p is a prime of the form $p = 5k + 2$ or $p = 5k + 3$ and investigate the structure of cyclic codes over $M_2(\mathbb{F}_p + u\mathbb{F}_p)$. Note that cyclic codes of length N were discussed for $p = 2$ in [15] along with the Gray map ϕ and some

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good parameters of linear codes over $\mathbb{F}_4$. It is worth mentioning that [15] determined the structure of cyclic codes only. Here, we extend this work for a class of primes satisfying $p \equiv 2$ or $3 (\text{mod } 5)$. Along with the structural properties of cyclic codes (in Section 3), we also discuss their dual codes and obtain conditions for such codes to be self-dual (in Section 4). Finally, in addition to few examples of self-dual cyclic codes over $\mathbb{F}_4$, we obtain many optimal and near to optimal linear codes over $\mathbb{F}_9$ in Table 1 and Table 2, respectively.

The presentation of the paper is as follows: In Section 2, we discuss some basic facts and establish an isometry between $M_2(\mathbb{F}_p + u\mathbb{F}_p)$ and $\mathbb{F}_{p^2} + u\mathbb{F}_{p^2} + v\mathbb{F}_{p^2} + uv\mathbb{F}_{p^2}$ where $u^2 = v^2 = 0, uv = vu$. In Section 3, by following the same arguments of [15], we determine the structure of cyclic codes while in Section 4, we study self-dual codes and obtain necessary and sufficient conditions for cyclic codes to be self-dual. Section 5 contains several optimal and near to optimal codes of length 16 and 20, respectively over $\mathbb{F}_9$. Section 6 concludes the paper.

2 Basic facts and Gray Map

Throughout, p denotes a prime such that $p \equiv 2$ or $3 (\text{mod } 5)$ and $\mathbb{F}_p$, a finite field of order p. Then from Lemma 4.1 of [5], we know that the polynomial $f(x) = x^2 + x + (p-1)$ is irreducible over $\mathbb{F}_p$. In fact, the irreducibility of $f(x)$ over $\mathbb{F}_p$ leads to restrict $p \equiv 2$ or $3 (\text{mod } 5)$. For example, if $p = 11$ satisfying $p \equiv 1 (\text{mod } 5)$, then $f(x) = x^2 + x + 10$ has roots 3, 7 and hence $f(x)$ is not an irreducible polynomial in $\mathbb{F}_p[x]$. One may take different irreducible polynomials over $\mathbb{F}_p$ for other primes p. But, here we fix the irreducible polynomial $f(x) = x^2 + x + (p-1)$, which has some specific benefit as discussed in the subsequent portion. Now, $\mathbb{F}_{p^2} \cong \mathbb{F}_p[w]$ is a Galois field of order p^2 where $f(w) = 0$, and the Galois group $G = \text{Gal}(\mathbb{F}_{p^2}/\mathbb{F}_p)$ is cyclic generated by the Frobenius automorphism $\sigma : \mathbb{F}_{p^2} \longrightarrow \mathbb{F}_{p^2}$ defined by $\sigma(a) = a^p$ for all $a \in \mathbb{F}_{p^2}$. Now, following the same arguments given in [16], we construct a cyclic algebra $\mathcal{D} = (\mathbb{F}_{p^2}/\mathbb{F}_p, \sigma, 1)$ as follows:

$\mathcal{D}$ contains the Galois field $\mathbb{F}_{p^2}$, and has a right $\mathbb{F}_{p^2}$-basis $\{1, e\}$ such that $\mathcal{D} = \mathbb{F}_{p^2} \oplus e\mathbb{F}_{p^2}$ where $ae = e\sigma(a)$ and $e^2 = 1$, for all $a \in \mathbb{F}_{p^2}$. Again, by ([18], Theorem 12.1, page-316), $\mathcal{D} = (\mathbb{F}_{p^2}/\mathbb{F}_p, \sigma, 1)$ is a central simple algebra over $\mathbb{F}_p$. Hence, $\mathcal{D}$ is a cyclic algebra and $\dim \mathcal{D} = 4$. Also, by Proposition 1 of [19], we have $\mathcal{D} \cong M_2(\mathbb{F}_p)$, i.e., $\mathbb{F}_{p^2} \oplus e\mathbb{F}_{p^2} \cong M_2(\mathbb{F}_p)$, where $M_2(\mathbb{F}_p)$ is the ring of 2×2 matrices over $\mathbb{F}_p$. Further, by considering $\mathbb{F}_{p^2}/\mathbb{F}_p$ as a vector space over $\mathbb{F}_p$, we have $\text{End}_{\mathbb{F}_p}(\mathbb{F}_{p^2}) \cong M_2(\mathbb{F}_p)$. Thus, $\mathcal{D} \cong \text{End}_{\mathbb{F}_p}(\mathbb{F}_{p^2})$, where the isomorphism $i : \mathcal{D} \longrightarrow \text{End}_{\mathbb{F}_p}(\mathbb{F}_{p^2})$ is given by $i(a) = l_a$, for all $a \in \mathbb{F}_{p^2}$ and $l_a : \mathbb{F}_{p^2} \longrightarrow \mathbb{F}_{p^2}$ is defined by $l_a(c) = ac$, for all $c \in \mathbb{F}_{p^2}$ and $i(e) = \sigma$. In fact, following [5] for $a = a_1 + wa_2 \in \mathbb{F}_{p^2}$ (with $a_1, a_2 \in \mathbb{F}_p$), we have $l_a \in \text{End}_{\mathbb{F}_p}(\mathbb{F}_{p^2})$ whose matrix representation is given by

$$l_a \longmapsto \begin{pmatrix} a_1 & a_2 \\ a_2 & a_1 + (p-1)a_2 \end{pmatrix} = L_a(\text{say}),$$

and also the matrix for e is

$$e \longmapsto \begin{pmatrix} 1 & 0 \\ p-1 & p-1 \end{pmatrix} = L_e(\text{say}).$$

To calculate L_e, we have used the fact $w^p \equiv (p-1)w + (p-1) \pmod{\text{w}^2 + \text{w} + (\text{p} - 1)}$ of ([5], Theorem 4.5). As pointed out above, this is one of the major reasons for the choice of irreducible polynomial $f(x) = x^2 + x + (p-1)$ over $\mathbb{F}_p$. In this way, $M_2(\mathbb{F}_p) \cong \mathbb{F}_{p^2} \oplus e\mathbb{F}_{p^2}$, where $ae = e\sigma(a)$, $e^2 = 1$ for all $a \in \mathbb{F}_{p^2}$, and the matrix correspondence is

Table 1 Cyclic codes of length 4 over $M_2(\mathbb{F}_3 + u\mathbb{F}_3)$

Generators of cyclic codes	$\phi(\mathcal{C})$	Remarks
$\langle f_2 f_3, (u + vw) f_4 \rangle$	[16, 11, 3]	near to optimal
$\langle uf_2, vf_3, (u + v) f_4 \rangle$	[16, 12, 2]	near to optimal
$\langle f_2, (u + v) f_3 \rangle$	[16, 14, 2]	optimal
$\langle f_3, uvf_4 \rangle$	[16, 13, 2]	near to optimal
$\langle f_3, vf_4 \rangle$	[16, 14, 2]	optimal
$\langle f_1, uf_2, vf_4 \rangle$	[16, 15, 2]	MDS
$\langle uf_2, vf_4, uvf_3 \rangle$	[16, 10, 4]	near to optimal

Table 2 Cyclic codes of length 5 over $M_2(\mathbb{F}_3 + u\mathbb{F}_3)$

Generators of cyclic codes	$\phi(\mathcal{C})$	Remarks
$\langle f_1, uf_2\rangle$	[20, 18, 2]	optimal
$\langle f_2, uvf_3\rangle$	[20, 13, 5]	near to optimal
$\langle f_3, uf_1\rangle$	[20, 16, 2]	near to optimal
$\langle f_2, uf_3, uvf_1f_2\rangle$	[20, 17, 3]	optimal
$\langle f_1, uf_2, vf_3\rangle$	[20, 19, 2]	MDS
$\langle f_1, (u+v)f_3\rangle$	[20, 18, 2]	optimal
$\langle f_1, uf_2, (u+v)f_3\rangle$	[20, 19, 2]	MDS

$$a + eb = (a_1 + wa_2 + e(b_1 + wb_2)) \longmapsto L_a + L_eL_b = \begin{pmatrix} a_1 + b_1 & a_2 + b_2 \\ a_2 - b_1 - b_2 & a_1 - a_2 - b_1 \end{pmatrix}.$$

Now, let $\xi = \alpha + e\beta \in \mathbb{F}_{p^2} \oplus e\mathbb{F}_{p^2}$ where $\alpha = \frac{p-1}{5} + w\frac{p-2}{5}, \beta = \frac{p-4}{5} + w\frac{2}{5} \in \mathbb{F}_{p^2}$. Consider $v = e + \xi$, then $v \in \mathbb{F}_{p^2} \oplus e\mathbb{F}_{p^2}$ and

$$\alpha^2 = \frac{1}{5}, \alpha^p = \frac{1+2w}{5}, \beta^p = -\frac{2}{5}(3+w),$$

and hence

$$\begin{aligned}
\xi^2 &= (\alpha + e\beta)(\alpha + e\beta) \\
&= \alpha^2 + \alpha e\beta + e\alpha\beta + e\beta e\beta \\
&= \alpha^2 + e\sigma(\alpha)\beta + e\alpha\beta + \sigma(\beta)\beta, (\text{since } e^2 = 1) \\
&= \alpha^2 + e\alpha^p\beta + e\alpha\beta + \beta^p\beta \\
&= \frac{1}{5} - e\frac{2w}{5} + e\frac{2w}{5} + \frac{4}{5} \\
&= 1.
\end{aligned}$$

Therefore, by using above values, we have

$$\begin{aligned}
v^2 &= (e + \xi)(e + \xi) \\
&= e^2 + e\xi + \xi e + \xi^2 \\
&= 1 + e(\alpha + e\beta) + (\alpha + e\beta)e + 1 \\
&= 2 + e\alpha + \beta + \alpha e + e\beta e, (\text{since } e^2 = 1) \\
&= 2 + e\alpha + \beta + e\sigma(\alpha) + \sigma(\beta), (\text{since } e^2 = 1) \\
&= 2 + e\alpha + \beta + e\alpha^p + \beta^p \\
&= 2 - \frac{e}{5}(1+2w) + \frac{2}{5}(-2+w) + \frac{e}{5}(1+2w) - \frac{2}{5}(3+w) \\
&= 2 - 2 = 0.
\end{aligned}$$

Also, the matrix representation of v is given by

$$v \longmapsto \begin{pmatrix} 1 & 0 \\ p-1 & p-1 \end{pmatrix} + \begin{pmatrix} p-1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ p-1 & 0 \end{pmatrix} = M_v(\text{say}) \in M_2(\mathbb{F}_p).$$

Hence, $M_v^2 = 0$. Again, as $v = e + \xi$ and $M_2(\mathbb{F}_p) \cong \mathbb{F}_{p^2} \oplus e\mathbb{F}_{p^2}$, we have $M_2(\mathbb{F}_p) \cong \mathbb{F}_{p^2} \oplus v\mathbb{F}_{p^2}$, where $v^2 = 0$.

Let $R = M_2(\mathbb{F}_p + u\mathbb{F}_p)$, $u^2 = 0$ be a noncommutative ring of 2×2 matrices over $\mathbb{F}_p + u\mathbb{F}_p$. Now, we can represent $R = M_2(\mathbb{F}_p) + uM_2(\mathbb{F}_p) \cong M_2(\mathbb{F}_p)[u]/\langle u^2\rangle$. Therefore, from above discussion, we have $R \cong \mathbb{F}_{p^2} + u\mathbb{F}_{p^2} + v\mathbb{F}_{p^2} + uv\mathbb{F}_{p^2}$, where $u^2 = v^2 = 0$, $uv = vu$. Hence, further we use R for both $\mathbb{F}_{p^2} + u\mathbb{F}_{p^2} + v\mathbb{F}_{p^2} + uv\mathbb{F}_{p^2}$ and $M_2(\mathbb{F}_p + u\mathbb{F}_p)$, where $u^2 = v^2 = 0$, $uv = vu$. Note that every element $\alpha \in R$ is of the form $\alpha = \alpha_1 + u\alpha_2 + v\alpha_3 + uv\alpha_4$ where $\alpha_i \in \mathbb{F}_{p^2}$, for $i = 1, 2, 3, 4$. Recall that a non-empty subset $\mathcal{C}$ of R^N is said to be a *linear code* if it is an R-submodule of R^N. Also, $\mathcal{C}$ is said to be a *cyclic code* if $(c_{N-1}, c_0, \ldots, c_{N-2}) \in \mathcal{C}$ for each $(c_0, c_1, \ldots, c_{N-1}) \in \mathcal{C}$. We consider the Gray map defined in [15] as follows:

$$\phi : R \longrightarrow \mathbb{F}_{p^2}^4$$

by

$$\phi(\alpha_1 + u\alpha_2 + v\alpha_3 + uv\alpha_4) = (\alpha_4, \alpha_3 + \alpha_4, \alpha_2 + \alpha_4, \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4). \tag{1}$$

Clearly, ϕ is a linear bijective map and can be extended to R^N componentwise. For any $\alpha = \alpha_1 + u\alpha_2 + v\alpha_3 + uv\alpha_4 \in R$, the Gray weight $wt_G(\alpha)$ is defined as $wt_G(\alpha) = wt_H(\alpha_4) + wt_H(\alpha_3 + \alpha_4) + wt_H(\alpha_2 + \alpha_4) + wt_H(\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4)$, where wt_H is the Hamming weight in $\mathbb{F}_{p^2}$. The Gray weight of $c = (c_0, c_1, \ldots, c_{N-1}) \in R^N$ is defined by $wt_G(c) = \sum_{i=0}^{N-1} wt_G(c_i)$ while the Gray distance of a linear code $\mathcal{C}$ is $d_G(\mathcal{C}) = \min\{wt_G(c) \mid 0 \neq c \in \mathcal{C}\}$. Therefore, it is a routine work to verify that ϕ is a weight preserving map from (R^N, Gray weight) to ($\mathbb{F}_{p^2}^{4N}$, Hamming weight).

Definition 1 [11, 12] Let v be a cyclic shift and $N = mt$, where m, t are positive integers. Then the quasi-cyclic shift operator π_t is defined by

$$\pi_t(c^1 \mid c^2 \mid \ldots \mid c^t) = (v(c^1) \mid v(c^2) \mid \ldots \mid v(c^t)),$$

where $c^i \in R^m$ for $i = 1, 2, \ldots, t$. A linear code $\mathcal{C}$ of length N over R is said to be a quasi-cyclic code of index t if and only if $\pi_t(\mathcal{C}) = \mathcal{C}$.

Theorem 1 *Let $\mathcal{C}$ be a cyclic code of length N, size M and distance d_G. Then $\phi(\mathcal{C})$ is a quasi-cyclic code of length $4N$, index 4, size M and distance d_H.*

Proof Note that ϕ is a distance preserving linear map. Therefore, for a cyclic code $\mathcal{C}$ of length N, size M and distance d_G, $\phi(\mathcal{C})$ is a linear code of length $4N$, size M and distance d_H. On the other hand, by Definition 1 and map ϕ, it is easy to check that $\phi v = \pi_4 \phi$. Now, for a cyclic code $\mathcal{C}$, we have $\phi(\mathcal{C}) = \phi v(\mathcal{C}) = \pi_4(\phi(\mathcal{C}))$. Hence, $\phi(\mathcal{C})$ is a quasi-cyclic code of length $4N$ and index 4 over $\mathbb{F}_{p^2}$. □

3 Cyclic codes over R

Let $\bar{\rho} : R \longrightarrow \mathbb{F}_{p^2}$ be a projection map defined by $\bar{\rho}(\alpha_1 + u\alpha_2 + v\alpha_3 + uv\alpha_4) = \alpha_1$, for $\alpha_i \in \mathbb{F}_{p^2}$, $i = 1, 2, 3, 4$. Then the ring homomorphism $\bar{\rho}$ can be extended to the polynomial ring $R[x]$ as $\rho : R[x] \longrightarrow \mathbb{F}_{p^2}[x]$ defined by $\rho(\sum_{i=0}^{N} \alpha_i x^i) = \sum_{i=0}^{N} \bar{\rho}(\alpha_i) x^i$. Now, $f(x)$ is called a basic irreducible polynomial in $R[x]$ if and only if $\rho(f(x))$ is irreducible in $\mathbb{F}_{p^2}[x]$.

Henceforth, we consider cyclic codes $\mathcal{C}$ of length N where p does not divide N, i. e., $\gcd(N, p) = 1$.

Theorem 2 *Let $\gcd(N, p) = 1$ and $x^N - 1 = \prod_{s=1}^{m} f_s$, where $f_1, f_2, \ldots, f_m$ are pairwise coprime and irreducible polynomials in $\mathbb{F}_{p^2}[x]$. Then $R[x]/\langle x^N - 1\rangle = \bigoplus_{s=1}^{m} R[x]/\langle f_s\rangle$.*

Proof It follows from the successful application of the Chinese Remainder Theorem by considering $R[x]/\langle x^N - 1\rangle$ as a right R-module (see [16, 20]). □

Proposition 1 *Let f be an irreducible polynomial in $\mathbb{F}_{p^2}[x]$. Then the possible right R-submodules of $R_f = R[x]/\langle f\rangle$ are $\langle 0\rangle, \langle 1\rangle, \langle u\rangle, \langle v\rangle, \langle uv\rangle, \langle u\rangle + \langle v\rangle$ and $\langle u + vH_f\rangle$ where H_f is a unit in $\mathbb{F}_{p^2}[x]$.*

Proof Let $I(\neq 0)$ be a right R-submodule of R_f and $a + \langle f\rangle$, a non-zero element of I. Then $a \notin \langle f\rangle$. Assume $a = a_1 + ua_2 + va_3 + uva_4$, where $a_i \in \mathbb{F}_{p^2}[x]$, for $i = 1, 2, 3, 4$.

Case A: If $\gcd(f, \rho(a)) = 1$, i.e., $\gcd(f, a_1) = 1$, then $\rho(a)$ is a unit in $\mathbb{F}_{p^2}[x]$, and hence a is unit in I. Thus, $I = \langle a\rangle = R_f$.

Case B: If $\gcd(f, \rho(a)) = f$, i.e., $\gcd(f, a_1) = f$, then

$$\begin{aligned} a + \langle f\rangle &= a_1 + ua_2 + va_3 + uva_4 + \langle f\rangle \\ &= ua_2 + va_3 + uva_4 + \langle f\rangle. \end{aligned} \tag{2}$$

Subcase I: If $\gcd(f, a_2) = f$, then from equation (2), we have

$$a + \langle f\rangle = va_3 + uva_4 + \langle f\rangle. \tag{3}$$

Therefore, $I \subseteq \langle v + \langle f \rangle \rangle$.

(1) If $\gcd(f, a_3) = 1$, then from equation (3), we have

$$uv = aa_3^{-1}u + \langle f \rangle \in I.$$

Again, by equation (3), we have

$$va_3 = a - uva_4 + \langle f \rangle \in I$$
$$\text{which implies } v = (a - uva_4)a_3^{-1} + \langle f \rangle \in I.$$

Hence, $\langle v + \langle f \rangle \rangle \subseteq I$. Thus, $I = \langle v + \langle f \rangle \rangle$.
(2) If $\gcd(f, a_3) = f$, then from equation (3),

$$a + \langle f \rangle = uva_4 + \langle f \rangle. \tag{4}$$

Therefore, $I \subseteq \langle uv + \langle f \rangle \rangle$. Again, since a is a non-zero element of I, so $\gcd(f, a_4) = 1$, and by equation (4), $uv = aa_4^{-1} + \langle f \rangle \in I$. Hence, $\langle uv + \langle f \rangle \rangle \subseteq I$ and $I = \langle uv + \langle f \rangle \rangle$.

Subcase II: Let $\gcd(f, a_2) = 1$.

If $\gcd(f, a_3) = f$, then from equation (2), we have

$$\begin{aligned} a + \langle f \rangle &= ua_2 + va_3 + uva_4 + \langle f \rangle \\ &= ua_2 + uva_4 + \langle f \rangle. \end{aligned} \tag{5}$$

Therefore, $I \subseteq \langle u + \langle f \rangle \rangle$. Again, by equation (5),

$$uv = aa_2^{-1}v + \langle f \rangle \in I.$$

Now, from equation (5),

$$u = (a - uva_4)a_2^{-1} + \langle f \rangle \in I.$$

Then $\langle u + \langle f \rangle \rangle \subseteq I$. Hence, $I = \langle u + \langle f \rangle \rangle$.
If $\gcd(f, a_3) = 1$, then by equation (2), we have

$$uv = aa_2^{-1}v + \langle f \rangle.$$

Again, by equation (2),

$$ua_2 + va_3 = a - uva_4 + \langle f \rangle \in I$$
$$\text{which implies } u + va_3a_2^{-1} = u + vH_f = (a - uva_4)a_3a_2^{-1} + \langle f \rangle \in I,$$

where $H_f = a_3a_2^{-1}$ is a unit in $\mathbb{F}_{p^2}[x]$. Then either $I = \langle u + vH_f \rangle$, or $\langle u + vH_f \rangle \subset I$. If $\langle u + vH_f \rangle \subset I$, then there exists $u + vH_{f'} \in I$ such that $u + vH_{f'} \notin \langle u + vH_f \rangle$ where $f' \in \mathbb{F}_{p^2}[x]$. Therefore, $v(H_f - H_{f'}) = u + vH_f - u - vH_{f'} \in I$. As $(H_f - H_{f'}) \notin \langle f \rangle$, so $v \in I$. Again, since a_2^{-1} exists, by equation (2), $uv \in I$. Further, from equation (2), $u \in I$ and this implies that $\langle u \rangle + \langle v \rangle \subseteq I$. Also, it is clear from equation (2) that $I \subseteq \langle u \rangle + \langle v \rangle$. Hence, $I = \langle u \rangle + \langle v \rangle$.
Thus, by combining above all cases, we conclude that the possible right R-submodules of $R_f = R[x]/\langle f \rangle$ are $\langle 0 \rangle, \langle 1 \rangle, \langle u \rangle, \langle v \rangle, \langle uv \rangle, \langle u \rangle + \langle v \rangle$ and $\langle u + vH_f \rangle$ where H_f is a unit in $\mathbb{F}_{p^2}[x]$. □

Proposition 2 *Let $x^N - 1 = \prod_{s=1}^{m} f_s$, where $f_1, f_2, \ldots, f_m$ are pairwise coprime and irreducible polynomials in $\mathbb{F}_{p^2}[x]$. Let $\hat{f_s} = \frac{x^N-1}{f_s}$, for $1 \le s \le m$. Then a right R-submodule of $R[x]/\langle x^N - 1 \rangle$ is the sum of right R-submodules $\langle \hat{f_s} + \langle x^N - 1 \rangle \rangle$, $\langle u\hat{f_s} + \langle x^N - 1 \rangle \rangle$, $\langle v\hat{f_s} + \langle x^N - 1 \rangle \rangle$, $\langle uv\hat{f_s} + \langle x^N - 1 \rangle \rangle$, $(\langle u\hat{f_s} \rangle + \langle v\hat{f_s} \rangle + \langle x^N - 1 \rangle)$ and $\langle (u + vH_f)\hat{f_s} + \langle x^N - 1 \rangle \rangle$ where H_f is a unit in $\mathbb{F}_{p^2}[x]$.*

Proof It follows from Theorem 2 and Proposition 1. □

Theorem 3 *Let $\mathcal{C}$ be a cyclic code of length N. Then*

$$\mathcal{C} = \langle \hat{F_1} \rangle \oplus \langle u\hat{F_2} \rangle \oplus \langle v\hat{F_3} \rangle \oplus \langle uv\hat{F_4} \rangle \oplus \langle (u + vH_f)\hat{F_5} \rangle \oplus (\langle u\hat{F_6} \rangle + \langle v\hat{F_6} \rangle)$$

and $|\mathcal{C}| = p^{8\deg F_1 + 4\deg F_2 + 4\deg F_3 + 2\deg F_4 + 4\deg F_5 + 6\deg F_6}$, where $F_0, F_1, \ldots, F_6$ are pairwise coprime and monic polynomials in $\mathbb{F}_{p^2}[x]$ such that $x^N - 1 = \prod_{s=0}^{6} F_s$ with $\hat{F_s} = \frac{x^N - 1}{F_s}$ and H_f is a unit in $\mathbb{F}_{p^2}[x]$.

Proof First part is same as the proof of ([15], Theorem 2).

Here, note that

$$|\langle \hat{F_1} \rangle| = p^{8\deg F_1}, |\langle u\hat{F_2} \rangle| = p^{4\deg F_2}, |\langle v\hat{F_3} \rangle| = p^{4\deg F_3}, |\langle uv\hat{F_4} \rangle| = p^{2\deg F_4},$$
$$|\langle (u + vH_f)\hat{F_5} \rangle| = p^{4\deg F_5}, |(\langle u\hat{F_6} \rangle + \langle v\hat{F_6} \rangle)| = p^{6\deg F_6}.$$

Therefore,

$|\mathcal{C}| = |\langle \hat{F_1} \rangle| \cdot |\langle u\hat{F_2} \rangle| \cdot |\langle v\hat{F_3} \rangle| \cdot |\langle uv\hat{F_4} \rangle| \cdot |\langle (u + vH_f)\hat{F_5} \rangle| \cdot |(\langle u\hat{F_6} \rangle + \langle v\hat{F_6} \rangle)| = p^{8\deg F_1 + 4\deg F_2 + 4\deg F_3 + 2\deg F_4 + 4\deg F_5 + 6\deg F_6}$. □

Proposition 3 *Let $\mathcal{C}$ be a cyclic code of length N over R. Then $\mathcal{C} = \langle A \rangle \oplus u\langle B \rangle \oplus v\langle C \rangle \oplus uv\langle D \rangle \oplus (u + vH_f)\langle F \rangle$ and $|\mathcal{C}| = p^{10N - 2(\deg A + \deg B + \deg C + \deg D + \deg F)}$, where A, B, C, D, F are polynomials in $\mathbb{F}_{p^2}[x]$ which are factors of $x^N - 1$.*

Proof Same as the proof of ([15], Proposition 4). □

Proposition 4 *Let $\mathcal{C}$ be a cyclic code of length N over R. Then there exists a polynomial F in R[x] such that $\mathcal{C} = \langle F \rangle$.*

Proof Let $\mathcal{C}$ be a cyclic code of length N over R. Then by Theorem 3, we have $\mathcal{C} = \langle \hat{F_1} \rangle \oplus \langle u\hat{F_2} \rangle \oplus \langle v\hat{F_3} \rangle \oplus \langle uv\hat{F_4} \rangle \oplus \langle (u + vH_f)\hat{F_5} \rangle \oplus (\langle u\hat{F_6} \rangle + \langle v\hat{F_6} \rangle)$, where $F_0, F_1, \ldots, F_6$ are pairwise coprime and monic polynomials in $\mathbb{F}_{p^2}[x]$ such that $x^N - 1 = \prod_{s=0}^{6} F_s$ with $\hat{F_s} = \frac{x^{N-1}}{F_s}$. Take $F = \hat{F_1} + u\hat{F_2} + v\hat{F_3} + uv\hat{F_4} + (u + vH_f)\hat{F_5} + (\langle u \rangle + \langle v \rangle)\hat{F_6}$. Now, we show $\mathcal{C} = \langle F \rangle$. It is easily seen that $\langle F \rangle \subseteq \mathcal{C}$. Also, for $s \neq k$, $x^N - 1$ is a factor of $\hat{F_s}\hat{F_k}$ and this implies that $\hat{F_s}\hat{F_k} = 0 (\text{mod } \mathrm{x^N} - 1)$, for $0 \leq s, k \leq 6$. Moreover, since polynomials $F_s, \hat{F_s}$ are coprime in $\mathbb{F}_{p^2}[x]$, there exist polynomials $g_s, h_s \in \mathbb{F}_{p^2}[x]$ such that $g_s F_s + h_s \hat{F_s} = 1$, for all $s = 1, 2, 3, 4, 5, 6$. Now, $\prod_{i=1}^{5}(g_s F_s + h_s \hat{F_s}) = 1$, and which implies that

$$\begin{aligned} &w_1 F_1 F_2 F_3 F_4 F_5 + w_2 \hat{F_1} F_2 F_3 F_4 F_5 + w_3 F_1 \hat{F_2} F_3 F_4 F_5 + w_4 F_1 F_2 \hat{F_3} F_4 F_5 \\ &+ w_5 F_1 F_2 F_3 \hat{F_4} F_5 + w_6 F_1 F_2 F_3 F_4 \hat{F_5} = 1, \end{aligned} \tag{6}$$

where $w_1, w_2, \ldots, w_6$ are some polynomials in $\mathbb{F}_{p^2}[x]$. Multiplying equation (6) by $\hat{F_6}$, we have

$$\hat{F_6} w_1 F_1 F_2 F_3 F_4 F_5 = \hat{F_6}.$$

Again,

$$\begin{aligned} F w_1 F_1 F_2 F_3 F_4 F_5 =& \{\hat{F_1} + u\hat{F_2} + v\hat{F_3} + uv\hat{F_4} + (u + vH_f)\hat{F_5} \\ &+ (\langle u \rangle + \langle v \rangle)\hat{F_6}\} w_1 F_1 F_2 F_3 F_4 F_5 \\ =& (\langle u \rangle + \langle v \rangle)\hat{F_6} w_1 F_1 F_2 F_3 F_4 F_5 \ (\text{since } F_s \hat{F_s} = 0, \forall 0 \leq s \leq 6) \\ =& (\langle u \rangle + \langle v \rangle)\hat{F_6}. \end{aligned}$$

Therefore, $(\langle u \rangle + \langle v \rangle)\hat{F_6} \in \langle F \rangle$. Again, by following similar procedure, we can show that $\hat{F_1}, u\hat{F_2}, v\hat{F_3}, uv\hat{F_4}, (u + vH_f)\hat{F_5} \in \langle F \rangle$. Hence, $\mathcal{C} \subseteq \langle F \rangle$. Thus, $\mathcal{C} = \langle F \rangle$. □

Corollary 1 *Every right R-submodule of $R[x]/\langle x^N - 1 \rangle$ is principally generated.*

4 Self-dual cyclic codes over *R*

Let $\mathcal{C}$ be a linear code of length N over R. For any two vectors $c = (c_0, c_1, \ldots, c_{N-1})$, $c' = (c'_0, c'_1, \ldots, c'_{N-1}) \in \mathcal{C}$, the Euclidean inner product is defined by $c \cdot c' = \sum_{i=0}^{N-1} c_i c'_i$. The dual of $\mathcal{C}$ is defined by $\mathcal{C}^{\perp} = \{x \in R^N \mid x \cdot y = 0, \forall y \in \mathcal{C}\}$. Recall that $\mathcal{C}$ is said to be *self-orthogonal* if $\mathcal{C} \subseteq \mathcal{C}^{\perp}$ and *self-*

dual if $\mathcal{C} = \mathcal{C}^{\perp}$. Let $f(x)$ be a polynomial such that $f(0) = a_0 \neq 0$. Then the reciprocal polynomial of $f(x)$ is denoted by $f^*(x)$, and defined as $f^*(x) = a_0^{-1}x^{\deg(f(x))}f(x^{-1})$. Note that $\deg(f(x)) = \deg(f^*(x))$. In the present section, we obtain the structure of dual codes (Theorem 4), and necessary and sufficient conditions (Theorem 5) for cyclic codes to be self-dual.

Theorem 4 *Let $\mathcal{C} = \langle \hat{F}_1\rangle \oplus \langle u\hat{F}_2\rangle \oplus \langle v\hat{F}_3\rangle \oplus \langle uv\hat{F}_4\rangle \oplus \langle (u+vH_f)\hat{F}_5\rangle \oplus (\langle u\hat{F}_6\rangle + \langle v\hat{F}_6\rangle)$ be a cyclic code of length N where $F_0, F_1, \ldots, F_6$ are pairwise coprime and monic polynomials in $\mathbb{F}_{p^2}[x]$ such that $x^N - 1 = \prod_{s=0}^{6} F_s$ with $\hat{F}_s = \frac{x^N-1}{F_s}$ and H_f is a unit in $\mathbb{F}_{p^2}[x]$. Then*

$$\mathcal{C}^{\perp} = \langle \hat{F}_0^*\rangle \oplus \langle u\hat{F}_2^*\rangle \oplus \langle v\hat{F}_3^*\rangle \oplus \langle uv\hat{F}_6^*\rangle \oplus \langle (u+vH_f)\hat{F}_5^*\rangle \oplus (\langle u\hat{F}_4^*\rangle + \langle v\hat{F}_4\rangle)$$

and $|\mathcal{C}^{\perp}| = p^{(8\deg F_0+4\deg F_2+4\deg F_3+6\deg F_4+4\deg F_5+2\deg F_6)}$.

Proof Let $\hat{\mathcal{C}} = \langle \hat{F}_0^*\rangle \oplus \langle u\hat{F}_2^*\rangle \oplus \langle v\hat{F}_3^*\rangle \oplus \langle uv\hat{F}_6^*\rangle \oplus \langle (u+vH_f)\hat{F}_5^*\rangle \oplus (\langle u\hat{F}_4^*\rangle + \langle v\hat{F}_4\rangle)$. Clearly, $\hat{\mathcal{C}} \subseteq \mathcal{C}^{\perp}$. Now, $|\mathcal{C}| = p^{8\deg F_1+4\deg F_2+4\deg F_3+2\deg F_4+4\deg F_5+6\deg F_6}$ and $|\mathcal{C}^{\perp}| = \frac{p^{8N}}{|\mathcal{C}|} = p^{(8\deg F_0+4\deg F_2+4\deg F_3+6\deg F_4+4\deg F_5+2\deg F_6)}$. Again,

$$|\langle \hat{F}_0^*\rangle| = p^{8\deg F_0}, |\langle u\hat{F}_2^*\rangle| = p^{4\deg F_2}, |\langle v\hat{F}_3^*\rangle| = p^{4\deg F_3}, |\langle uv\hat{F}_6^*\rangle| = p^{2\deg F_6},$$
$$|\langle (u+vH_f)\hat{F}_5^*\rangle| = p^{4\deg F_5}, |\langle (\langle u\hat{F}_4^*\rangle + \langle v\hat{F}_4\rangle)\rangle| = p^{6\deg F_4},$$

and

$$|\hat{\mathcal{C}}| = p^{(8\deg F_0+4\deg F_2+4\deg F_3+6\deg F_4+4\deg F_5+2\deg F_6)}.$$

Hence, $\hat{\mathcal{C}} = \mathcal{C}^{\perp}$. □

Proposition 5 *Let $\mathcal{C}$ be a cyclic code of length N over R. Then $\mathcal{C}^{\perp} = \langle A^*\rangle \oplus u\langle B^*\rangle \oplus v\langle C^*\rangle \oplus uv\langle D^*\rangle \oplus (u+vH_f)\langle F^*\rangle$ and $|\mathcal{C}| = p^{10N-2(\deg A^*+\deg B^*+\deg C^*+\deg D^*+\deg F^*)}$, where A^*, B^*, C^*, D^*, F^* are polynomials in $\mathbb{F}_{p^2}[x]$ which are factors of $x^N - 1$.*

Proof Same as the proof of Proposition 3. □

Proposition 6 *Let $\mathcal{C}$ be a cyclic code of length N over R. Then there exists a polynomial F^* in $R[x]$ such that $\mathcal{C}^{\perp} = \langle F^*\rangle$.*

Proof Consider $F^* = \hat{F}_0^* + u\hat{F}_2^* + v\hat{F}_3^* + uv\hat{F}_6^* + (u+vH_f)\hat{F}_5^* + (\langle u\rangle + \langle v\rangle)\hat{F}_4^*$ and follows the same argument of Proposition 4. □

Theorem 5 *Let $\mathcal{C} = \langle \hat{F}_1\rangle \oplus \langle u\hat{F}_2\rangle \oplus \langle v\hat{F}_3\rangle \oplus \langle uv\hat{F}_4\rangle \oplus \langle (u+vH_f)\hat{F}_5\rangle \oplus (\langle u\hat{F}_6\rangle + \langle v\hat{F}_6\rangle)$ be a cyclic code of length N where $F_0, F_1, \ldots, F_6$ are pairwise coprime and monic polynomials in $\mathbb{F}_{p^2}[x]$ such that $x^N - 1 = \prod_{s=0}^{6} F_s$ with $\hat{F}_s = \frac{x^N-1}{F_s}$ and H_f is a unit in $\mathbb{F}_{p^2}[x]$. Then $\mathcal{C}$ is a self-dual cyclic code if and only if $F_1 = F_0^*, F_2 = F_2^*, F_3 = F_3^*, F_4 = F_6^*, F_5 = F_5^*, F_6 = F_4^*$.*

Proof It is known that the code $\mathcal{C} = \langle \hat{F}\rangle$ is self-dual if and only if $\hat{F} = \hat{F}^*$, i.e., $\hat{F}_1 = \hat{F}_0^*, \hat{F}_2 = \hat{F}_2^*, \hat{F}_3 = \hat{F}_3^*, \hat{F}_4 = \hat{F}_6^*, \hat{F}_5 = \hat{F}_5^*, \hat{F}_6 = \hat{F}_4^*$, and this implies that $F_1 = F_0^*, F_2 = F_2^*, F_3 = F_3^*, F_4 = F_6^*, F_5 = F_5^*, F_6 = F_4^*$. □

Proposition 7 *Let $\mathcal{C} = \langle A\rangle \oplus u\langle B\rangle \oplus v\langle C\rangle \oplus uv\langle D\rangle \oplus (u+vH_f)\langle F\rangle$ be a cyclic code of length N over R, where A, B, C, D, F are polynomials in $\mathbb{F}_{p^2}[x]$ which are factors of $x^N - 1$. Then $\mathcal{C}$ is a self-dual cyclic code if and only if $A = A^*, B = B^*, C = C^*, D = D^*, F = F^*$.*

5 Examples

It is well known that a linear code $[n,k,d]_q$ satisfies the Singleton bound $d \leq n-k+1$, and is called an MDS (maximum-distance-separable) code if $d = n-k+1$. The best-known linear codes (close to MDS) are catalogued in the online database [6], which provides both bounds and known linear codes over $\mathbb{F}_q$ where $q = p^n \leq 9$. Recall that a linear code $[n,k,d]_q$ is said to be *optimal* (resp. *near to optimal*) if there exists a code $[n,k,d']_q$ in [6] such that $d' - d = 0$ (resp. $d' - d \leq 2$). An optimal code need not necessarily be an

MDS code. For example, in Table 1, the code $[16, 15, 2]_9$ is both optimal and MDS whereas $[16, 14, 2]_9$ is optimal but not MDS. In this section, by using the database [6], we present several optimal and near to optimal codes of length 16 and 20, respectively over $\mathbb{F}_9$ under the Gray images of cyclic codes over R. All computations involved in these examples are carried out by using the Magma computation system [3].

Example 1 Let $\mathbb{F}_4 = \{0, 1, w, w+1\}$ be the field of order 4 where $w^2 + w + 1 = 0$. Now, in $\mathbb{F}_4[x]$, we have

$$x^5 - 1 = (x+1)(x^2 + wx + 1)(x^2 + wx + x + 1).$$

Let $f_1 = x + 1, f_2 = x^2 + wx + 1$ and $f_3 = x^2 + wx + x + 1$. Then $f_1^* = f_1, f_2^* = f_2, f_3^* = f_3$. Hence, $\mathcal{C}_1 = \langle u\hat{f_2}, v\hat{f_3}, uv\hat{f_1}, (u+v)\hat{f_1}\rangle$ and $\mathcal{C}_2 = \langle u\hat{f_1}, v\hat{f_3}, uv\hat{f_2}, (u+v)\hat{f_2}\rangle$ are self-dual cyclic codes of length 5 over $R = M_2(\mathbb{F}_2 + u\mathbb{F}_2)$. Moreover, $\phi(\mathcal{C}_i)$ is a [20, 10, 4] linear code over $\mathbb{F}_4$, for $i = 1, 2$.

Example 2 Let $\mathbb{F}_9 = \mathbb{F}_3[w]$ be the field of order 9 where $w^2 + w + 2 = 0$. Then, in $\mathbb{F}_9[x]$,

$$x^4 - 1 = (x+1)(x+2)(x+1+2w)(x+2+w).$$

Let $f_1 = x + 1, f_2 = x + 2, f_3 = x + 1 + 2w$ and $f_4 = x + 2 + w$. Now, some optimal and good codes of length 16 over $\mathbb{F}_9$ are enlisted in Table 1.

Example 3 Let $\mathbb{F}_4 = \{0, 1, w, w+1\}$ be the field of order 4 where $w^2 + w + 1 = 0$. Now, in $\mathbb{F}_4[x]$,

$$x^3 - 1 = (x+1)(x+w)(x+w+1).$$

Let $f_1 = x + 1, f_2 = x + w$ and $f_3 = x + w + 1$. Then $f_1^* = f_1, f_2^* = f_3, f_2^* = f_2$. Therefore, $\mathcal{C} = \langle u\hat{f_1}, v\hat{f_1}, uv\hat{f_2}, (u+v)\hat{f_3}\rangle$ is a self-dual cyclic code of length 3 over $M_2(\mathbb{F}_2 + u\mathbb{F}_2)$ and $\phi(\mathcal{C})$ is a [12, 6, 4] linear code over $\mathbb{F}_4$.

Example 4 Let $\mathbb{F}_9 = \mathbb{F}_3[w]$ be the field of order 9 where $w^2 + w + 2 = 0$. Now, in $\mathbb{F}_9[x]$, we have

$$x^5 - 1 = (x+2)\{x^2 + (w+1)x + 1\}\{x^2 + 2wx + 1\}.$$

Let $f_1 = x + 2, f_2 = x^2 + (w+1)x + 1$ and $f_3 = x^2 + 2wx + 1$. Now, we present some cyclic codes of length 5 over $M_2(\mathbb{F}_3 + u\mathbb{F}_3)$ and their Gray images in Table 2.

6 Conclusion

In this article, we studied cyclic, and in particular self-dual cyclic codes over $R = M_2(\mathbb{F}_p + u\mathbb{F}_p)$. We obtained necessary and sufficient conditions for cyclic codes to be self-dual over R. It is proved that cyclic codes of length N with $\gcd(N, p) = 1$ are principally generated. As an application, we presented some optimal and good codes of length 16 and 20 over $\mathbb{F}_9$, respectively. We believe that some more new and optimal codes of different length can be obtained by considering their Gray images in our set up.

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